

Lecture 06 Dependency, range bounds, and subdivision

Prerequisites: Lectures 04–05; elementary derivatives and Python loops.

Goal: explain wide enclosures, improve them without losing coverage, and distinguish a bound from a proof of a root.

1. A valid enclosure need not be a sharp range

0–13 minutes: definitions and a diagnostic example

For a real function f and an interval X contained in its domain, the *range* is the set $f(X) = \{f(x) : x \in X\}$. An interval evaluator F is an *inclusion function* when

$$f(X) \subseteq F(X).$$

The left side describes the mathematical problem; the right side is what our algorithm returns. A continuous function on a closed, bounded interval has a minimum and a maximum, so its range is itself a closed interval. Computing those two extrema exactly can nevertheless be difficult.

Replacing each operation in an expression by a valid interval operation gives a *natural interval extension*. The inclusion proof follows the expression tree: each intermediate interval contains every intermediate real value. This proves validity, but it does not prove that every number in the final interval occurs as a function value. Wider intermediate sets can introduce combinations of values that no single input can produce.

For example, take $f(x) = x^2 + 1$ on $X = [-1, 1]$. Independent interval multiplication gives $X \cdot X = [-1, 1]$, and hence $X \cdot X + 1 = [0, 2]$. The sharp square operation instead gives $\text{sqr}(X) + 1 = [1, 2]$, the exact range. Both enclosures are valid. Only the second immediately proves that f is strictly positive.

Reading a certificate. If $0 \notin F(X)$, then f has no zero in X . If $0 \in F(X)$, the conclusion is only “this enclosure did not exclude a zero.” Existence requires another argument, such as a sign change combined with continuity. A nonnegative lower endpoint proves nonnegativity; a *strictly* positive lower endpoint proves positivity.

A useful additional property. An evaluator is *inclusion isotonic* if $X \subseteq Y$ implies $F(X) \subseteq F(Y)$. Exact natural interval extensions have this property wherever all operations are defined. It explains why restricting the input can help. Validity alone does not imply isotonicity, and changing the implementation or precision while comparing outputs requires care: two valid enclosures need not be nested.

Discuss before calculating

An implementation returns $[-100, 100]$ for a function whose true range is $[2, 3]$. Is it incorrect? Is it useful for excluding a zero? What changes if the result is $[2.1, 3]$? Identify the missing value that makes the latter fail.

Our objective is to preserve inclusion while making the enclosure useful for a specific question. Small width is helpful, but it is not the definition of correctness. Even the exact range can be wide when the function varies greatly.

2. One function, three different computations

13–30 minutes: work through the algebra together

Consider $f(x) = x(1 - x)$ on $X = [0, 1]$. The identities

$$x - x^2 = x(1 - x) = \frac{1}{4} - \left(x - \frac{1}{2}\right)^2$$

hold for every real x . Their interval evaluations differ because interval operations retain ranges of intermediate values, not the relations between those values. Use a sharp square whenever a square is written explicitly.

Expression	Interval calculation on $[0, 1]$	Enclosure
$x - x^2$	$[0, 1] - [0, 1]$	$[-1, 1]$
$x(1 - x)$	$[0, 1] \cdot [0, 1]$	$[0, 1]$
$\frac{1}{4} - (x - \frac{1}{2})^2$	$\frac{1}{4} - [0, \frac{1}{4}]$	$[0, \frac{1}{4}]$

The last enclosure is the exact range. Indeed, the square is nonnegative, so $f(x) \leq 1/4$; also $x \geq 0$ and $1 - x \geq 0$, so $f(x) \geq 0$. The values $f(0) = 0$ and $f(1/2) = 1/4$ attain the bounds. Continuity supplies every intermediate value. Equivalently, $f'(x) = 1 - 2x$ changes sign at $1/2$.

Where does the false upper bound 1 come from? Introduce $y = 1 - x$. The exact problem maximizes xy along the line segment $x + y = 1$ inside the square $[0, 1]^2$. Independent multiplication maximizes xy over the whole square. It permits $(x, y) = (1, 1)$ even though that pair violates $y = 1 - x$. The computation solved a larger problem, whose answer is still an upper bound for the original one.

This loss of information is called the *dependency problem*. It can occur even with exact rational endpoints: none of the three calculations in the table uses rounding. Increasing machine precision therefore cannot remove these particular excess widths. Precision addresses arithmetic rounding; rewriting an expression addresses which relations the evaluator preserves.

There is no universal rule that the expanded or factored form is best. The input interval matters. On $[0, 1/4]$, the three forms above give, in order, $[-1/16, 1/4]$, $[0, 1/4]$, and $[0, 3/16]$. The vertex form remains exact here, but finding a comparable identity for a complicated function may be difficult. Subdivision offers a systematic alternative that does not require a perfect symbolic reformulation.

Board exercise: trace the repeated variable

In $X - X^2$, which choices of the two occurrences of x produce the interval lower bound -1 ? Can one real input make both choices at once? Explain why the bound is valid despite being unattainable.

Practical rule. Before adding precision or splitting hundreds of intervals, inspect the formula. A sharp square, an exact identity, or an analytically known sign may save substantial computation. Check the domain whenever a rewriting introduces division or another restricted operation.

3. Covering an interval gives a proof

30–46 minutes: the covering argument and two subintervals

Suppose closed intervals $X_1, \dots, X_N$ cover the entire input interval: $X = \bigcup_{i=1}^N X_i$. Each piece must lie in the domain of the evaluator. If $f(X_i) \subseteq F(X_i)$ for every piece, then

$$f(X) = \bigcup_{i=1}^N f(X_i) \subseteq \bigcup_{i=1}^N F(X_i) \subseteq \text{hull}\{F(X_1), \dots, F(X_N)\}.$$

To prove this, take an arbitrary $x \in X$. Coverage puts x in at least one piece X_i . The local inclusion puts $f(x)$ in $F(X_i)$, and therefore in the hull. Since x was arbitrary, no input is left out. This is the step that a finite list of point samples cannot supply.

For sorted pieces of $[a, b]$, check the two outer endpoints and check that successive pieces touch or overlap. Shared endpoints cause no problem. A small gap does: it is a region about which the local certificates say nothing. Exact rational split points make these coverage checks transparent.

For $0 \leq a \leq b \leq 1$, the product expression for our example gives

$$F([a, b]) = [a, b] [1 - b, 1 - a] = [a(1 - b), b(1 - a)].$$

All factors are nonnegative, which explains the simple endpoint formula. It is an enclosure, not generally the exact range on the piece. Splitting $[0, 1]$ into $[0, 1/2]$ and $[1/2, 1]$ gives $[0, 1/2]$ on each piece. Their hull is $[0, 1/2]$, an improvement over $[0, 1]$ that still contains $[0, 1/4]$.

```
from fractions import Fraction

def product_enclosure(left, right):
    assert 0 <= left <= right <= 1
    return left * (1 - right), right * (1 - left)

def uniform_pieces(left, right, count):
    # Inputs left and right are Fractions; count is an integer.
    if count < 1 or left > right:
        raise ValueError("Need ordered endpoints and count >= 1")
    step = (right - left) / count
    pieces = []
    for index in range(count):
        a = left + index * step
        b = left + (index + 1) * step
        pieces.append((a, b))
    return pieces
```

These short functions use exact arithmetic. The assertions document the example's restricted domain; they are not a general interval library. The same covering argument works with outward-rounded machine intervals, provided each local evaluation has a valid inclusion guarantee.

A logical checkpoint

If every piece has a strictly positive lower bound, what follows globally? Would the same conclusion follow if all piece midpoints had positive values? State the extra information supplied by the interval evaluations.

4. Implement the hull and inspect the improvement

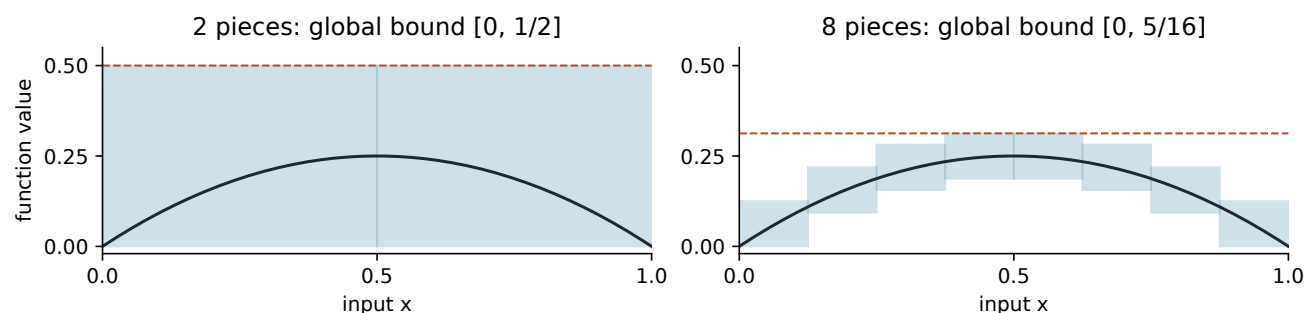
46–62 minutes: readable code and a geometric picture

A global enclosure uses the smallest local lower endpoint and the largest local upper endpoint. This combines certificates; averaging endpoints would not preserve the inclusion argument. The function below assumes a nonempty list of covering pieces and an evaluator valid on every piece.

```
def enclose_by_pieces(evaluate, pieces):
    left, right = pieces[0]
    lower, upper = evaluate(left, right)
    for left, right in pieces[1:]:
        local_lower, local_upper = evaluate(left, right)
        lower = min(lower, local_lower)
        upper = max(upper, local_upper)
    return lower, upper

for count in [1, 2, 4, 8, 16]:
    pieces = uniform_pieces(Fraction(0), Fraction(1), count)
    lower, upper = enclose_by_pieces(product_enclosure, pieces)
    print(count, str(lower), str(upper))
```

The output upper endpoints are 1, 1/2, 3/8, 5/16, and 9/32; all lower endpoints are zero. Each vertical rectangle below encloses every point of the function graph over its horizontal input piece. The drawn curve is illustrative: the endpoint arithmetic and coverage establish the bounds.



For an even number N of equal pieces, put $h = 1/N$. The upper endpoint on the piece starting at $a = kh$ is

$$(a + h)(1 - a) = \frac{(1 + h)^2}{4} - \left(a - \frac{1 - h}{2}\right)^2.$$

The nearest allowed starting points to $(1 - h)/2$ are $h/2$ away when N is even. Thus the global upper bound is $1/4 + h/2 = 1/4 + 1/(2N)$. Halving the mesh halves the excess upper bound in this particular example. This is a derived result for this evaluator, not a general convergence rate.

What does refinement converge to?

Predict the result for $N = 32$. Does the width approach zero or 1/4? Explain why a nonzero limiting width is compatible with a perfect enclosure.

5. Choose the right remedy for a wide bound

62–77 minutes: precision, subdivision, and elementary functions

Width can reflect three different things: variation of the true function, lost dependence between intermediate quantities, and outward rounding. Only the last is directly controlled by arithmetic precision. Subdivision reduces the input variation on each piece and often reduces dependency loss. The global hull must still include the full variation across all pieces. For our polynomial, a global width below $1/4$ would be an error.

If a fixed evaluator is inclusion isotonic, each child enclosure is contained in its parent's enclosure. Replacing a parent by a covering collection of children then cannot enlarge the global hull. Strict improvement is not guaranteed. Nor does validity by itself imply that repeated subdivision converges to the exact range; that requires properties of the evaluator.

Stopping honestly. A maximum number of pieces is a resource limit, not a proof of a requested accuracy. If the budget is exhausted, report the current bound and the unmet target. In an adaptive method, unprocessed pieces must retain valid bounds so that coverage is never silently lost. Likewise, splitting an interval containing zero does not make $1/x$ defined at zero. Domain failure requires reformulating the question or reporting it.

For elementary functions, use an implementation with inclusion guarantees. Evaluating an ordinary `math.sin` at two endpoints is insufficient: even exact endpoint values for $[0, \pi]$ would miss the interior maximum 1. The course uses Arb through `python-flint`. Here is a direct example, independent of the teaching package `vc`:

```
from flint import arb, ctx, fmpq

with ctx.workprec(100):
    x = arb(0).union(arb(fmpq(1, 2)))
    result = (x.sin() - x * x + 1) * x.cos()
    if not result.is_finite():
        raise ValueError("No finite enclosure was obtained")
    midpoint = result.mid().fmpq()
    radius = result.rad().fmpq()
    lower = midpoint - radius
    assert lower > 0
```

The input ball contains $[0, 1/2]$. Arb's operations enclose the expression on that ball; subtracting its exact rational radius from its exact rational midpoint gives a rigorous lower endpoint. The successful comparison proves $(\sin x - x^2 + 1) \cos x > 0$ for every $x \in [0, 1/2]$. A conversion to a ball may widen the domain; all operations must be valid on that enlarged input too.

Explain the scope of the computation

Which line supplies coverage of the input? Which supplies the sign certificate? Why would a positive printed midpoint alone be insufficient? If the assertion failed, would that prove that the function has a zero?

Reading and notebook. The elementary-function example accompanies Notebook/Lab 06. API reference: [python-flint](#), [real balls \(arb\)](#).

6. Exercises, checks, and a certificate audit

77–90 minutes: select two exercises; finish the others independently

1. **Formula choice.** On $X = [0, 1/4]$, evaluate $X - X^2$, $X(1 - X)$, and $1/4 - (X - 1/2)^2$ using sharp squares. Verify the three bounds on page 2. Use monotonicity of $x(1 - x)$ on this interval to find its exact range. Which forms introduce unattainable values, and where?
2. **Prove positivity.** Let $g(x) = x^2 - x + 1$ on $[0, 1]$. The natural expanded expression gives $[0, 2]$. Explain precisely why that result does not prove strict positivity. Rewrite the expression to obtain the exact range and a positive lower bound. Would higher precision alone improve the exact-arithmetic expanded enclosure?
3. **The missing piece.** A program checks only $[0, 49/100]$ and $[51/100, 1]$ for $q(x) = (x - 1/2)^2$. It obtains a strictly positive lower bound on each piece and declares no roots on $[0, 1]$. Find the error and the missed root. Repair the cover, and predict why the repaired calculation can no longer certify strict positivity.
4. **Set a meaningful budget.** For the product evaluator and equal subdivision of $[0, 1]$, use the upper-bound formula for even N . What is the smallest power of two N for which the upper bound exceeds the true maximum by at most $1/100$? Could any valid subdivision produce a global enclosure of width at most $1/100$? Distinguish these two targets.
5. **Validity versus isotonicity (optional).** Let $f(x) = 0$ for all real x . Define $F(X) = [-1, 1]$ if the width of X is less than 1, and $F(X) = [0, 0]$ otherwise. Show that F is always an inclusion function. Use $X = [0, 1/2] \subset Y = [0, 2]$ to disprove isotonicity. What happens to the enclosure when Y is split into four equal pieces?

Selected checks. Exercise 1: the bounds are $[-1/16, 1/4]$, $[0, 1/4]$, and $[0, 3/16]$; the last is the exact range since $f'(x) = 1 - 2x \geq 1/2$ on this input. Exercise 2: $g(x) = (x - 1/2)^2 + 3/4$ gives $[3/4, 1]$. Exercise 3: the missed root is $1/2$; each checked piece has exact lower bound $1/10000$, but together they do not cover the input. Adding the middle piece includes a zero value. Exercise 4: $1/(2N) \leq 1/100$ requires $N \geq 50$, so the smallest power of two is 64. The actual range has width $1/4$, ruling out the second target. Exercise 5: the parent returns $[0, 0]$, while the children's hull is $[-1, 1]$. Both are valid, but subdivision worsened the result.

Certificate audit: say all the premises

Complete this claim: “The function is positive on the whole interval because...” Your explanation must identify the exact function and domain, the covering pieces, a valid local evaluator, and the strict sign of every local lower bound. State what a failed check leaves undecided.

Before the next lecture. Run the exact-rational subdivision example and retain the endpoint fractions. Try a different formula on each piece, keeping coverage fixed. Notebook 06 provides further experiments; Lecture 07 adds derivative information to obtain sharper bounds and faster certificates.

Reading and notebook. Tucker, *Validated Numerics*, §3.1 (pp. 46–54), especially Examples 3.1.1–3.1.4 and Theorem 3.8. These notes use original derivations; the numbered notebook and lab provide the executable course companion.